Lecture 10

Peter–Weyl and completeness

Objectives. *Lecture 5 proved that matrix coefficients of inequivalent irreducible representations are orthogonal. This lecture proves that finite-dimensional matrix coefficients span a dense subspace of the square-integrable functions on a compact Lie group. It then deduces that every compact Lie group is a closed subgroup of a unitary group.*

By the end of this lecture you will be able to:

- Average a rank-one projection to obtain a nonzero positive compact intertwiner, and use it to prove that the finite-dimensional invariant subspaces span a dense subspace in every strongly continuous unitary representation of a compact Lie group.
- Identify the finite-dimensional subspaces of $L^2(K)$ invariant under the right translations R with spaces of matrix coefficients, and state the Peter–Weyl theorem with multiplicities.
- Prove that a compact Lie group has a faithful finite-dimensional unitary representation.
- Compute the coefficient blocks of $\mathrm{U}(1)$ and $\mathrm{SU}(2)$.

Two results from earlier courses are special cases of the theorem of this lecture. For a finite group G, the regular module gives $\mathbb{C}[G] \cong \bigoplus_U \mathrm{End}(U)$, so that $\sum_U (\dim U)^2 = |G|$ (Finite Groups). For the circle $\mathrm{U}(1)$, a square-integrable function has a Fourier series in the functions $e^{in\theta}$. The Peter–Weyl theorem contains both. Each irreducible representation Π of a compact Lie group K gives a block of d_Π^2 matrix coefficients, and the span of these blocks is dense in $L^2(K)$. For $\mathrm{SU}(2)$ the blocks have $1, 4, 9, \dots$ functions. Compared with a finite group, the new difficulty is the infinite sum.

The proof has three parts. First we average a rank-one projection over the group. The result commutes with the group, and it is a compact operator. Next, its eigenspaces for nonzero eigenvalues are finite-dimensional and invariant, so every unitary representation contains finite-dimensional pieces. Finally we apply this to $L^2(K)$, where a finite-dimensional invariant piece turns out to consist of matrix coefficients.

10.1 Analytic preliminaries

Throughout, K is a compact Lie group. We write $\int_K f(g)\,dg$ for its normalised Haar integral on $C(K)$. We import four facts from Lecture 5.

(H1) The integral is left and right invariant. It is invariant under $g \mapsto g^{-1}$, and $\int_K 1\,dg = 1$.

(H2) If $f \in C(K)$ is nonnegative and not identically zero, then $\int_K f\,dg > 0$.

(H3) A finite-dimensional continuous representation has an invariant inner product. It is then an orthogonal sum of irreducible subrepresentations.

(H4) Schur orthogonality, in the form (10.1) below.

We also import Fubini's theorem for continuous functions on $K \times K$ from Bröcker–tom Dieck, Section I.5. It states that the two iterated Haar integrals agree.

Let (Π, V) be a finite-dimensional continuous unitary representation. Its *matrix coefficients* are the continuous functions

$$c^{\Pi}_{u,v}(x) = \langle u, \Pi(x)v\rangle, \qquad u, v \in V.$$

The *coefficient space* $\mathscr{R}(K)$ is the span of all such functions, for all such (Π, V). Let $\mathrm{Irr}(K)$ be a set of representatives of the isomorphism classes of irreducible finite-dimensional continuous unitary representations. Write $d_\Pi = \dim V_\Pi$. For $\Pi, \Pi' \in \mathrm{Irr}(K)$, Schur orthogonality (H4) reads

$$\int_K \overline{c^{\Pi}_{u,v}(x)}\, c^{\Pi'}_{u',v'}(x)\,dx = \begin{cases} d_\Pi^{-1}\langle u', u\rangle\langle v, v'\rangle, & \Pi' = \Pi, \\ 0, & \Pi' \neq \Pi. \end{cases} \tag{10.1}$$

Definition 10.1. Let $(H_i)_{i\in I}$ be mutually orthogonal closed subspaces of a Hilbert space H. Their *Hilbert sum* $\widehat{\bigoplus}_i H_i$ is the closure of the algebraic sum $\bigoplus_i H_i$.

The elements of the Hilbert sum are the convergent series $\sum_i v_i$ with $v_i \in H_i$ and $\sum_i \|v_i\|^2 < \infty$. For Hilbert spaces H_i not given inside one space, the *external Hilbert sum* $\widehat{\bigoplus}_i H_i$ is the space of families (v_i) with $v_i \in H_i$ and $\sum_i \|v_i\|^2 < \infty$.

A bounded operator on a Hilbert space H is *compact* if it is an operator-norm limit of finite-rank operators. A self-adjoint operator T on H is *positive* if $\langle v, Tv\rangle \geq 0$ for all $v \in H$.

We import the spectral theorem for compact self-adjoint operators from Linear Algebra for Physics, Weeks 13–14, in the following form. If T is compact and self-adjoint, then

$$H = \ker T \;\widehat{\oplus}\; \widehat{\bigoplus_{\mu\neq 0}} \ker(T - \mu).$$

Moreover, each $\ker(T-\mu)$ with $\mu \neq 0$ is finite-dimensional. If $T \neq 0$, then T has a nonzero eigenvalue.

Definition 10.2. The space $L^2(K)$ is the Hilbert completion of $C(K)$ for the sesquilinear form $\langle f, h\rangle = \int_K \overline{f(g)}\, h(g)\,dg$.

By positivity (H2), this sesquilinear form is an inner product on $C(K)$. Hence the map $C(K) \to L^2(K)$ is injective. An element of $L^2(K)$ is *continuous* if it lies in $C(K)$. Only continuous elements have values $f(x)$. For continuous functions, $\|f\|_2 \leq \|f\|_\infty = \sup_x |f(x)|$.

Definition 10.3. A *strongly continuous unitary representation* of K on a Hilbert space H is a homomorphism Π from K to the unitary group of H such that $g \mapsto \Pi(g)v$ is continuous for every $v \in H$.

We do not ask for continuity in operator norm. Already right translation on $L^2(\mathrm{U}(1))$ is not norm continuous (Exercise 10.2).

Lemma 10.4. *Let $F\colon K \to H$ be continuous. There is a unique vector $\int_K F(g)\,dg$ in H with*

$$\left\langle w, \int_K F(g)\,dg \right\rangle = \int_K \langle w, F(g)\rangle\,dg \qquad (w \in H).$$

Moreover $\|\int_K F\| \le \int_K \|F(g)\|\,dg$, and $A\int_K F = \int_K AF$ for bounded A. If each $F(g)$ lies in a closed subspace W, so does $\int_K F$. The integral is unchanged if $F(g)$ is replaced by $F(hg)$.

Proof. The scalar integrand is continuous. The map $w \mapsto \overline{\int_K \langle w, F\rangle}$ is linear with bound $\|w\| \int_K \|F\|$. The Riesz representation theorem for Hilbert spaces, from Linear Algebra for Physics, gives the vector and the norm bound. For bounded A, $\langle w, A\int F\rangle = \int \langle A^* w, F\rangle = \int \langle w, AF\rangle$. If $w \perp W$, the integrand vanishes, so $\int F \in W^{\perp\perp} = W$. Invariance follows from the left invariance in (H1). ∎

10.2 An averaged compact operator

Theorem 10.5. *Let Π be a strongly continuous unitary representation of K on $H \neq 0$.*

(a) The space H contains a nonzero finite-dimensional invariant subspace.

(b) The sum H_{fin} of all finite-dimensional invariant subspaces is dense in H.

We look for a nonzero compact self-adjoint intertwiner. An intertwiner has invariant eigenspaces. A compact self-adjoint operator has finite-dimensional eigenspaces for nonzero eigenvalues. Averaging a rank-one projection over K gives an intertwiner, and compactness of K keeps it a compact operator.

Proof. Fix a unit vector $u \in H$. Let E_u be the rank-one projection given by $E_u v = \langle u, v\rangle u$. Then $\Pi(g)E_u\Pi(g)^{-1}v = \langle \Pi(g)u, v\rangle \Pi(g)u$, which is continuous in g. Define the average

$$Tv = \int_K \langle \Pi(g)u, v\rangle\, \Pi(g)u\,dg. \tag{10.2}$$

Step 1: T is bounded, positive and nonzero. The vector-integral lemma, Lemma 10.4, gives $\|Tv\| \le \|v\|$ and

$$\langle w, Tv\rangle = \int_K \langle \Pi(g)u, v\rangle \langle w, \Pi(g)u\rangle\,dg.$$

Exchanging v, w and conjugating gives the same integral, so $T^* = T$. For $w = v$ the integrand is $|\langle \Pi(g)u, v\rangle|^2 \ge 0$, so T is positive. For $v = w = u$ this integrand is continuous with value 1 at e. By positivity (H2), $\langle u, Tu\rangle > 0$. Hence T is nonzero.

Step 2: T commutes with Π. For $h \in K$,

$$T\Pi(h)v = \int_K \langle \Pi(h^{-1}g)u, v\rangle\, \Pi(g)u\,dg = \int_K \langle \Pi(g)u, v\rangle\, \Pi(h)\Pi(g)u\,dg = \Pi(h)Tv.$$

The first equality uses unitarity. The second replaces g by hg. The third takes $A = \Pi(h)$ in the vector-integral lemma, Lemma 10.4.

Step 3: T is compact. The orbit $O = \{\Pi(g)u : g \in K\}$ is compact, since it is a continuous image of K. Let $\varepsilon > 0$. Since O is compact, finitely many open balls of radius ε with centres in O cover O.

Let P be the orthogonal projection onto the span of the centres. Put $x_g = \Pi(g)u$ and $y_g = Px_g$. Each x_g lies within ε of some centre, and y_g is the nearest point of the span of the centres to x_g. Hence $\|x_g - y_g\| < \varepsilon$ for all g. Now

$$\langle x_g, v\rangle\langle w, x_g\rangle - \langle y_g, v\rangle\langle w, y_g\rangle = \langle x_g - y_g, v\rangle\langle w, x_g\rangle + \langle y_g, v\rangle\langle w, x_g - y_g\rangle.$$

Because $\|x_g\| = 1$ and $\|y_g\| \le 1$, the Cauchy–Schwarz inequality shows that this difference has modulus at most $2\varepsilon\|v\|\|w\|$. On the other hand,

$$\langle w, PTPv\rangle = \langle Pw, TPv\rangle = \int_K \langle y_g, v\rangle\langle w, y_g\rangle \, dg.$$

Subtracting gives $|\langle w, (T - PTP)v\rangle| \le 2\varepsilon\|v\|\|w\|$, so $\|T - PTP\| \le 2\varepsilon$. The operator PTP has finite rank, since P has finite rank. Hence T is compact.

Step 4: proof of (a). By Steps 1 and 3, T is compact, self-adjoint and nonzero. Hence the spectral theorem gives an eigenvalue $\mu \neq 0$ of T with finite-dimensional eigenspace $H_\mu \neq 0$. If $Tv = \mu v$, then Step 2 gives $T\Pi(h)v = \Pi(h)Tv = \mu\Pi(h)v$. So H_μ is invariant.

Step 5: proof of (b). A sum of two finite-dimensional invariant subspaces is again one, so H_{fin} is a subspace. Its closure M is invariant. Hence $M^\perp$ is closed and invariant, since $\langle \Pi(g)w, m\rangle = \langle w, \Pi(g)^{-1}m\rangle$. Restricting Π gives a strongly continuous unitary representation on $M^\perp$. If $M^\perp \neq 0$, the existence statement (a), applied to $M^\perp$, gives a finite-dimensional invariant subspace $W \neq 0$ in $M^\perp$. But $W \subset H_{\mathrm{fin}} \subset M$. Thus $W \subset M \cap M^\perp = 0$, a contradiction. ∎

Corollary 10.6. *Let Π be a strongly continuous unitary representation of K on H. Suppose that Π is irreducible, meaning that $H \neq 0$ and that the only closed invariant subspaces are 0 and H. Then* $\dim H < \infty$.

Proof. Since $H \neq 0$, the existence statement, part (a) of Theorem 10.5, gives a nonzero finite-dimensional invariant subspace W. Finite-dimensional subspaces are closed, so $W = H$ by irreducibility. ∎

This corollary uses compactness. A noncompact Lie group can have irreducible unitary representations of infinite dimension, and Part II treats them.

10.3 The regular representation

We begin with a tool for continuous functions on K, which we use throughout this section.

Lemma 10.7. *(a) For $\varphi \in C(K)$ and $\varepsilon > 0$ there is an open $U \ni e$ with $|\varphi(yx) - \varphi(x)| < \varepsilon$ and $|\varphi(xy) - \varphi(x)| < \varepsilon$ for all $x \in K$, $y \in U$.*

(b) For each open $U \ni e$ there is a continuous $\varphi \ge 0$ vanishing outside U with $\varphi(e) > 0$ and $\int_K \varphi = 1$.

Proof. (a) The function $F(y,x) = |\varphi(yx) - \varphi(x)| + |\varphi(xy) - \varphi(x)|$ is continuous on $K \times K$ and vanishes on $\{e\} \times K$. For each x there are open sets $U_x \ni e$ and $N_x \ni x$ with $F < \varepsilon$ on $U_x \times N_x$. Finitely many $N_{x_1}, \dots, N_{x_r}$ cover K. Take $U = \bigcap_j U_{x_j}$.

(b) Fix a norm $|\cdot|$ on $\mathfrak{k}$, and let B_r be the open ball of radius r about 0. By Lecture 2, exp maps B_r homeomorphically onto an open neighbourhood of e for all sufficiently small $r > 0$. Shrink r so that $\exp(B_r) \subset U$. Put $\psi(\exp X) = \max(0, r/2 - |X|)$ on $\exp(B_r)$ and $\psi = 0$ elsewhere. Then ψ is continuous on $\exp(B_r)$.

The closed ball $\overline{B}_{r/2}$ is compact, so its image under exp is compact and hence closed in the Hausdorff space K. The function ψ vanishes on the open complement $K \setminus \exp(\overline{B}_{r/2})$. The two open sets $\exp(B_r)$ and $K \setminus \exp(\overline{B}_{r/2})$ cover K, so ψ is continuous on K. Since $\psi(e) = r/2 > 0$, positivity (H2) gives $\int \psi > 0$. Take $\varphi = \psi / \int \psi$. Then $\varphi(e) > 0$. ∎

For $f \in C(K)$ define

$$(L_g f)(x) = f(g^{-1}x), \qquad (R_g f)(x) = f(xg).$$

Lemma 10.8. *The translations L and R extend to commuting strongly continuous unitary representations on $L^2(K)$.*

Proof. We treat R. The proof for L is the same, except that it uses left invariance and replaces U below by $U \cap U^{-1}$.

First, R is a homomorphism on $C(K)$. For $f \in C(K)$ and $g, h, x \in K$,

$$(R_g R_h f)(x) = (R_h f)(xg) = f(xgh) = (R_{gh} f)(x).$$

By the right invariance in (H1), R_g preserves $\|\cdot\|_2$. So R_g extends to an isometry of $L^2(K)$ with inverse $R_{g^{-1}}$.

Next, we prove strong continuity at e. Let $f \in L^2(K)$ and $\varepsilon > 0$. Choose $f_0 \in C(K)$ with $\|f - f_0\|_2 < \varepsilon$. Uniform continuity, Lemma 10.7(a), gives an open $U \ni e$ with $\|R_g f_0 - f_0\|_2 \le \|R_g f_0 - f_0\|_\infty < \varepsilon$ for $g \in U$. For $g \in U$, the triangle inequality and the isometry property give

$$\|R_g f - f\|_2 \le \|R_g(f - f_0)\|_2 + \|R_g f_0 - f_0\|_2 + \|f_0 - f\|_2 < 3\varepsilon.$$

Continuity at g_0 follows from $R_g f - R_{g_0} f = R_{g_0}\big(R_{g_0^{-1}g} f - f\big)$, since R_{g_0} is an isometry.

Finally, for $f \in C(K)$ both $L_g R_h f$ and $R_h L_g f$ are the function $x \mapsto f(g^{-1}xh)$. The two bounded operators therefore agree on the dense subspace $C(K)$, so they commute on $L^2(K)$. ∎

For $\varphi \in C(K)$ and $f \in L^2(K)$ put

$$R_\varphi f = \int_K \varphi(g) R_g f \, dg.$$

Elements of $L^2(K)$ have no values at points, but a matrix coefficient must have them. Averaging removes this difficulty. The element $R_\varphi f$ is a weighted average of right translates of f, and it is continuous.

Lemma 10.9 (Smoothing). *For $\varphi \in C(K)$ and $f \in L^2(K)$, the element $R_\varphi f$ is continuous, and*

$$(R_\varphi f)(x) = \langle k_x, f \rangle, \qquad k_x(h) = \overline{\varphi(x^{-1}h)}.$$

Proof. Put $(Sf)(x) = \langle k_x, f\rangle$. Fix $\varepsilon > 0$. Choose U for φ by uniform continuity, Lemma 10.7(a). If $x' \in xU^{-1}$, then $y = x'^{-1}x \in U$. Writing $t = x^{-1}h$ gives $|k_x(h) - k_{x'}(h)| = |\varphi(t) - \varphi(yt)| < \varepsilon$. By the Cauchy–Schwarz inequality, $|Sf(x) - Sf(x')| \le \varepsilon\|f\|_2$. So $Sf \in C(K)$ and $\|Sf\|_2 \le \|Sf\|_\infty \le \|\varphi\|_\infty\|f\|_2$. For $f, w \in C(K)$, Fubini and the substitution $h = xg$ give

$$\langle w, R_\varphi f\rangle = \int_K \varphi(g) \int_K \overline{w(x)} f(xg)\,dx\,dg = \int_K \overline{w(x)} \int_K \varphi(x^{-1}h) f(h)\,dh\,dx = \langle w, Sf\rangle.$$

As $C(K)$ is dense, $R_\varphi f = Sf$ for $f \in C(K)$. Both operators are bounded, so they agree on $L^2(K)$. ∎

Proposition 10.10. *(a) Let $W \subset L^2(K)$ be a finite-dimensional R-invariant subspace. Then $W \subset C(K)$. There is $u_W \in W$ with $f(x) = \langle u_W, R_x f\rangle$ for all $f \in W$ and $x \in K$.*

(b) The coefficient space $\mathscr{R}(K)$ is the sum of all finite-dimensional R-invariant subspaces. It is dense in $L^2(K)$.

Proof. (a) Choose an orthonormal basis $f_1, \dots, f_n$ of W. Strong continuity gives an open $U \ni e$ with $\|R_g f_i - f_i\| < 1/(2n)$ for all i and $g \in U$. By Lemma 10.7(b), there is a continuous $\varphi \ge 0$ vanishing outside U with $\int_K \varphi = 1$. This normalisation gives $R_\varphi f - f = \int_K \varphi(g)\big(R_g f - f\big)\,dg$. For a unit vector $f = \sum a_i f_i$ we have $|a_i| \le 1$. Hence

$$\|R_\varphi f - f\| \le \sum_i |a_i| \int_K \varphi(g)\|R_g f_i - f_i\|\,dg \le n \cdot \frac{1}{2n} = \frac{1}{2}.$$

So $\|R_\varphi f\| \ge \tfrac12$ for every unit vector $f \in W$, and R_φ is injective on W. Since W is closed and R-invariant, the vector-integral lemma, Lemma 10.4, shows that R_φ maps W into W. As W is finite-dimensional, $R_\varphi W = W$. Every element of W is therefore of the form $R_\varphi f$, and the smoothing lemma, Lemma 10.9, gives $W \subset C(K)$.

Now evaluation $f \mapsto f(e)$ is a linear functional on W. The Riesz representation theorem in W gives u_W with $f(e) = \langle u_W, f\rangle$. For $f \in W$, $R_x f$ lies in W and is the function $h \mapsto f(hx)$. Hence $f(x) = (R_x f)(e) = \langle u_W, R_x f\rangle$.

(b) The restriction of R to W is a finite-dimensional continuous unitary representation. By the reproducing formula in (a), each $f \in W$ is a matrix coefficient of this representation. Hence $W \subset \mathscr{R}(K)$. Conversely, $R_h c^{\Pi}_{u,w} = c^{\Pi}_{u,\Pi(h)w}$, so each $c^{\Pi}_{u,v}$ lies in the finite-dimensional R-invariant space $\{c^{\Pi}_{u,w} : w \in V\}$. Density follows from the density statement Theorem 10.5(b), applied to the representation R of the regular-representation lemma, Lemma 10.8. ∎

10.4 The Peter–Weyl theorem

The group $K \times K$ acts on $L^2(K)$ by

$$((g,h)\cdot f)(x) = f(g^{-1}xh), \qquad \text{that is, } (g,h) \mapsto L_g R_h.$$

For $\Pi \in \operatorname{Irr}(K)$, give $\operatorname{End}(V_\Pi)$ the Hilbert–Schmidt inner product $\langle A, B\rangle_{\mathrm{HS}} = \operatorname{tr}(A^* B)$ and the action $(g,h)\cdot A = \Pi(h) A \Pi(g)^{-1}$.

Let $\overline{V}_\Pi$ be the conjugate space, with $\overline{\Pi}(g)\bar{u} = \overline{\Pi(g)u}$ and inner product $\langle \bar{u}, \bar{v}\rangle = \langle v, u\rangle$. For this inner product, $\overline{\Pi}$ is unitary. The representation $\overline{\Pi}$ is irreducible, since $W \mapsto \overline{W}$ matches the invariant subspaces of Π and $\overline{\Pi}$. The coefficients of $\overline{\Pi}$ are the conjugate functions:

$$c^{\overline{\Pi}}_{\bar{u},\bar{v}}(x) = \big\langle \bar{u}, \overline{\Pi(x)v}\big\rangle = \langle \Pi(x)v, u\rangle = \overline{c^{\Pi}_{u,v}(x)}.$$

The space $\overline{V}_\Pi$ is isomorphic to V_Π^* by $\bar{u} \mapsto \langle u, \cdot\rangle$. The map $\bar{u} \otimes v \mapsto v\langle u, \cdot\rangle$ identifies $\overline{V}_\Pi \otimes V_\Pi$ with $\mathrm{End}(V_\Pi)$. For $g, h \in K$, unitarity of $\Pi(g)$ gives $\Pi(h)v\langle u, \Pi(g)^{-1}\cdot\rangle = \Pi(h)v\langle \Pi(g)u, \cdot\rangle$. The left side is the action of (g,h) on $v\langle u,\cdot\rangle$. Hence the pair (g,h) acts on $\overline{V}_\Pi \otimes V_\Pi$ as $\overline{\Pi}(g) \otimes \Pi(h)$. Thus the first tensor factor carries the left action L.

In Dirac notation, $v\langle u,\cdot\rangle$ is the operator $|v\rangle\langle u|$, and $c^\Pi_{u,v}(x) = \langle u|\Pi(x)|v\rangle$ is a matrix element. Under (g,h) the ket moves by $\Pi(h)$ and the bra by $\Pi(g)$, and bras form the conjugate space $\overline{V}_\Pi$.

We define

$$\Phi_\Pi \colon \mathrm{End}(V_\Pi) \to C(K), \qquad \Phi_\Pi(A)(x) = \sqrt{d_\Pi}\,\operatorname{tr}\big(\Pi(x)A\big),$$

and let E_Π be its image. For $A = v\langle u,\cdot\rangle$ we get $\Phi_\Pi(A) = \sqrt{d_\Pi}\, c^\Pi_{u,v}$.

Fix an orthonormal basis $e_1, \dots, e_{d_\Pi}$ of V_Π, and write $\Pi_{ij} = c^\Pi_{e_i,e_j}$. This is the entry of the matrix of $\Pi(x)$ in row i and column j. The operator $e_j\langle e_i,\cdot\rangle$ has matrix with a single entry 1, in row j and column i. Since $\operatorname{tr}\big(\Pi(x)e_j\langle e_i,\cdot\rangle\big) = \Pi_{ij}(x)$, the map Φ_Π sends this operator to $\sqrt{d_\Pi}\,\Pi_{ij}$. So a column of the matrix of A corresponds to a row of the array (Π_{ij}).

Theorem 10.11 (Peter–Weyl). *(a) Each Φ_Π is a $K \times K$-equivariant isometry from $(\mathrm{End}\, V_\Pi, \mathrm{HS})$ onto E_Π, and $E_\Pi \perp E_{\Pi'}$ for $\Pi \neq \Pi'$.*

(b) The coefficient space satisfies $\mathscr{R}(K) = \bigoplus_{\Pi \in \mathrm{Irr}(K)} E_\Pi$, an algebraic direct sum.

(c) The Hilbert completion of $\mathscr{R}(K)$ is $L^2(K) = \widehat{\bigoplus}_{\Pi\in\mathrm{Irr}(K)} E_\Pi$. The map $\bigoplus_\Pi \Phi_\Pi$ extends to a unitary $K \times K$-isomorphism

$$\widehat{\bigoplus_\Pi}\,\mathrm{End}(V_\Pi) \cong \widehat{\bigoplus_\Pi}\,\overline{V}_\Pi \otimes V_\Pi \longrightarrow L^2(K).$$

(d) Under R, $E_\Pi \cong V_\Pi^{\oplus d_\Pi}$ and $\dim \mathrm{Hom}_K(V_\Pi, L^2(K)) = d_\Pi$. Under L, $E_\Pi \cong \overline{V}_\Pi^{\oplus d_\Pi}$, and again each irreducible occurs with multiplicity equal to its dimension.

The theorem says that functions on K are organised in square blocks, one for each irreducible representation. In an orthonormal basis, block Π holds the d_Π^2 matrix entries Π_{ij}, up to the factor $\sqrt{d_\Pi}$. Figure 10.1 draws these blocks as arrays of functions. Since $\Pi(xh) = \Pi(x)\Pi(h)$, we have $\Pi_{ij}(xh) = \sum_k \Pi_{ik}(x)\Pi_{kj}(h)$, so R_h keeps the row i. Since $\Pi(g^{-1}x) = \Pi(g^{-1})\Pi(x)$, we have $\Pi_{ij}(g^{-1}x) = \sum_k \Pi_{ik}(g^{-1})\Pi_{kj}(x)$, so L_g keeps the column j.

Proof. (a) For $A = v\langle u,\cdot\rangle$ and $B = v'\langle u',\cdot\rangle$ we have $A^*B = u\langle v, v'\rangle\langle u',\cdot\rangle$, so $\operatorname{tr}(A^*B) = \langle u', u\rangle\langle v, v'\rangle$. By Schur orthogonality (10.1),

$$\langle \Phi_\Pi(A), \Phi_\Pi(B)\rangle = d_\Pi \cdot d_\Pi^{-1}\langle u', u\rangle\langle v, v'\rangle = \langle A, B\rangle_{\mathrm{HS}}.$$

Both sides are sesquilinear and rank-one operators span $\mathrm{End}(V_\Pi)$. The same argument gives $E_\Pi \perp E_{\Pi'}$. Equivariance is the calculation

$$\Phi_\Pi(A)(g^{-1}xh) = \sqrt{d_\Pi}\operatorname{tr}\big(\Pi(g)^{-1}\Pi(x)\Pi(h)A\big) = \Phi_\Pi\big(\Pi(h)A\Pi(g)^{-1}\big)(x).$$

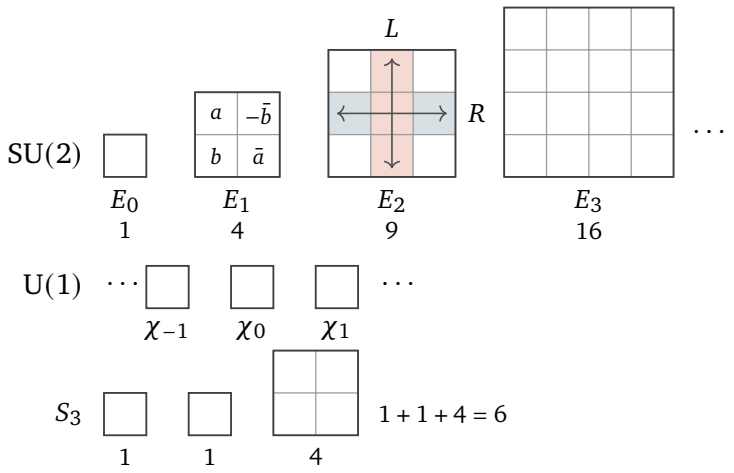


Figure 10.1: Blocks of functions in $L^2(K)$ for SU(2), U(1) and the finite group S_3. A cell is one entry Π_{ij}, with row i and column j, and a number below a block counts its cells. The SU(2) block E_1 shows the entries of the defining representation. In the block E_2, right translation R mixes the entries of the shaded row and left translation L mixes the entries of the shaded column.

(b) Let (σ, V) be a finite-dimensional continuous unitary representation. By complete reducibility (H3), $V = \bigoplus_j V_j$ orthogonally with each V_j irreducible. Write $u = \sum u_j$ and $v = \sum v_j$. Since $\sigma(x)v_k \in V_k$, we get $c^{\sigma}_{u,v} = \sum_j c^{\sigma_j}_{u_j,v_j}$.

Suppose that $\sigma_j \cong \Pi \in \mathrm{Irr}(K)$, and let S be an isomorphism. Since σ_j and Π are unitary, S^* is also an intertwiner. By Schur's lemma, $S^*S = \kappa \cdot 1$, and $\kappa > 0$ because S is injective. Hence $U = \kappa^{-1/2} S$ is a unitary intertwiner, and $c^{\sigma_j}_{u_j,v_j} = c^{\Pi}_{Uu_j,Uv_j} \in E_\Pi$. Thus $\mathscr{R}(K) = \sum_\Pi E_\Pi$. The sum is direct by the orthogonality in (a).

(c) Each E_Π is finite-dimensional, hence closed. By density of coefficients, Proposition 10.10(b), their sum is dense. So $L^2(K)$ is their Hilbert sum. The map $\bigoplus \Phi_\Pi$ is isometric with dense image, so it extends to a unitary. Equivariance passes to the closure by continuity.

(d) Under R alone, $A \mapsto \Pi(h)A$ acts on each column of A, so $E_\Pi \cong V_\Pi^{\oplus d_\Pi}$. Let $J : V_\Pi \to L^2(K)$ intertwine R. By the Hilbert-sum decomposition (c), $Jv = \sum_{\Pi'} P_{\Pi'} Jv$ in $L^2(K)$. The orthogonal projection $P_{\Pi'}$ onto $E_{\Pi'}$ commutes with R, because $E_{\Pi'}$ and $E_{\Pi'}^{\perp}$ are invariant. For $\Pi' \neq \Pi$, Schur's lemma gives $P_{\Pi'} J = 0$. So J maps into E_Π, and $\dim \mathrm{Hom}_K(V_\Pi, L^2(K)) = \dim \mathrm{Hom}_K(V_\Pi, V_\Pi^{\oplus d_\Pi}) = d_\Pi$.

Under L, $A \mapsto A\Pi(g)^{-1}$ acts on rows, giving $\overline{V}_\Pi^{\oplus d_\Pi}$. If $\overline{\Pi} \cong \overline{\Pi'}$, then conjugating the intertwiner gives $\Pi \cong \Pi'$. Also $\overline{\overline{\Pi}} = \Pi$. So conjugation induces a dimension-preserving bijection of the isomorphism classes in $\mathrm{Irr}(K)$. The projections $P_{\Pi'}$ also commute with L, so the same Hilbert-sum and Schur arguments apply. ∎

Theorem 10.12. *The coefficient space $\mathscr{R}(K)$ is dense in $C(K)$ for $\|\cdot\|_\infty$.*

Proof. The constant 1 is a coefficient of the trivial representation. Products are coefficients, since $c^{\Pi}_{u,v} c^{\Pi'}_{u',v'} = c^{\Pi\otimes\Pi'}_{u\otimes u', v\otimes v'}$. Conjugates are coefficients, since $\overline{c^{\Pi}_{u,v}} = c^{\overline{\Pi}}_{\bar{u},\bar{v}}$ for the inner product

on $\overline{V}_\Pi$ fixed above. Hence $\mathscr{R}(K)$ is a subalgebra of $C(K)$ that contains the constants and is closed under conjugation. It remains to prove that $\mathscr{R}(K)$ separates points.

Let $x \neq y$ and suppose $f(x) = f(y)$ for all $f \in \mathscr{R}(K)$. Since $L_g c^\Pi_{u,v} = c^\Pi_{\Pi(g)u,v}$, the space $\mathscr{R}(K)$ is L-invariant, so $f(g^{-1}x) = f(g^{-1}y)$ for all g. Put $s = x^{-1}y \neq e$ and $g = xt^{-1}$. Then $f(t) = f(ts)$ for all t. Hence $R_s f = f$ on $\mathscr{R}(K)$. Since $\mathscr{R}(K)$ is dense in $L^2(K)$ by Proposition 10.10(b), and R_s is bounded, $R_s = 1$ on $L^2(K)$.

Since K is Hausdorff, some open $U \ni e$ omits s. By Lemma 10.7(b), some continuous $\varphi \geq 0$ vanishes outside U and satisfies $\varphi(e) > 0 = \varphi(s)$. Then $(R_s\varphi)(e) = \varphi(s) \neq \varphi(e)$, so $R_s\varphi \neq \varphi$ in $C(K)$. Since $C(K) \to L^2(K)$ is injective, $R_s\varphi \neq \varphi$ in $L^2(K)$, which contradicts $R_s = 1$. Hence $\mathscr{R}(K)$ separates points.

We import the complex Stone–Weierstrass theorem. For compact Hausdorff X, it states that a subalgebra of $C(X)$ is uniformly dense if it contains the constants, is closed under conjugation and separates points. We apply it to $\mathscr{R}(K) \subset C(K)$. ∎

Separation of points has a structural consequence. A compact Lie group is defined abstractly, but it is always a group of matrices.

Theorem 10.13 (Faithful representation)**.** *A compact Lie group K has an injective finite-dimensional continuous unitary representation $\sigma\colon K \to \mathrm{U}(n)$. Its image is a closed subgroup of $\mathrm{U}(n)$, and σ is an isomorphism of Lie groups from K onto this image.*

Proof. We add summands to a representation until its kernel is trivial, and we show that this process stops.

Step 1: a smaller kernel. Let (σ, V) be a finite-dimensional continuous unitary representation with kernel N. For example, the trivial representation on $\mathbb{C}$ has kernel K. The kernel N is a closed subgroup of K, so it is a compact Lie group by the closed-subgroup theorem of Lecture 3. Suppose $N \neq \{e\}$, and choose $s \in N$ with $s \neq e$. By the proof of Theorem 10.12, the functions in $\mathscr{R}(K)$ separate e and s. These functions are spanned by matrix coefficients. So some finite-dimensional continuous unitary representation Π has a matrix coefficient with $c^\Pi_{u,v}(s) \neq c^\Pi_{u,v}(e)$. Hence $\Pi(s) \neq I$. The direct sum $\sigma \oplus \Pi$ has kernel $N' = N \cap \ker \Pi$. This is a closed subgroup of N, and $N' \neq N$ because $s \notin N'$.

Step 2: comparing N' with N. Since $N' \subseteq N$, we have $\dim N' \leq \dim N$. Suppose $\dim N' = \dim N$. The inclusion $N' \to N$ is an injective immersion between manifolds of equal dimension, so it is a local diffeomorphism. Hence N' is open in N. Let N_0 be the identity component of N. Then $N' \cap N_0$ is an open subgroup of the connected group N_0. An open subgroup is also closed, since its complement is a union of its cosets. So $N' \cap N_0 = N_0$, that is, $N_0 \subseteq N'$. The connected set N_0 is the largest connected subset of N containing e, so it is also the identity component of N'.

The components of N are the cosets of N_0. They are open and they cover the compact group N, so there are finitely many. Since $N_0 \subseteq N'$ and $N' \neq N$, the group N' consists of fewer of these cosets. So N' has fewer components than N.

Step 3: the process stops. By Step 2, each application of Step 1 either lowers the dimension of the kernel, or keeps the dimension and lowers the finite number of components. The dimension can decrease only finitely often. Between two decreases, the number of components decreases at each step. So after finitely many steps the kernel is $\{e\}$.

Step 4: the image. We now have an injective representation, and an orthonormal basis of its space gives $\sigma\colon K \to \mathrm{U}(n)$. The image $\sigma(K)$ is compact, so it is closed in $\mathrm{U}(n)$, and it is a Lie group by the closed-subgroup theorem. A continuous bijection from a compact space onto a Hausdorff space is a homeomorphism. So σ and its inverse are continuous homomorphisms between Lie groups. By the automatic smoothness theorem of Lecture 3, both are smooth. ∎

The proof applies to disconnected K. Step 3 uses that K is a Lie group. Compactness is also needed, since a covering group of a matrix group need not be a matrix group (Lecture 4).

10.5 Examples

Example 10.14 (The circle group)**.** Let $K = \mathrm{U}(1)$. Here $\int f = \frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta})\,d\theta$. Irreducible representations of an abelian group are one-dimensional by Schur's lemma. By Lecture 7 they are $\chi_n(e^{i\theta}) = e^{in\theta}$, $n \in \mathbb{Z}$. Then $d_{\chi_n} = 1$ and $E_n = \mathbb{C}\chi_n$. Hence

$$\mathscr{R}(\mathrm{U}(1)) = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}\chi_n, \qquad L^2(\mathrm{U}(1)) = \widehat{\bigoplus}_{n\in\mathbb{Z}} \mathbb{C}\chi_n.$$

Every $f \in L^2$ equals $\sum_n \hat{c}_n\chi_n$ with $\hat{c}_n = \langle \chi_n, f\rangle$ and $\|f\|_2^2 = \sum |\hat{c}_n|^2$. This is L^2 completeness of Fourier series.

For the $K \times K$ action,

$$((g,h)\cdot\chi_n)(x) = \chi_n(g^{-1}xh) = \overline{\chi_n(g)}\,\chi_n(h)\,\chi_n(x).$$

So $\mathbb{C}\chi_n \cong \overline{\chi}_n \otimes \chi_n$. Under L the line $\mathbb{C}\chi_n$ carries χ_{-n}. Under R it carries χ_n.

We can now see the operator (10.2) in this case. Let $\Pi = R$ on $L^2(\mathrm{U}(1))$, and let u be a unit vector with coefficients $a_n = \langle \chi_n, u\rangle$. Then $\langle \chi_k, R_{e^{it}}u\rangle = e^{ikt}a_k$. Hence

$$\langle \chi_k, T\chi_m\rangle = \frac{1}{2\pi}\int_0^{2\pi} e^{ikt}a_k\,\overline{e^{imt}a_m}\,dt = \delta_{km}|a_m|^2.$$

So $T\chi_m = |a_m|^2\chi_m$. Since $\sum_n |a_n|^2 = 1$, for each $\mu > 0$ only finitely many n have $|a_n|^2 = \mu$. So each eigenspace of T with $\mu \neq 0$ is finite-dimensional, as the spectral theorem says.

Example 10.15 (The group SU(2))**.** Let $K = \mathrm{SU}(2)$ and write

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

By Lecture 8, $\mathrm{Irr}(\mathrm{SU}(2)) = \{V_m : m \geq 0\}$, where V_m has dimension $m+1$. Realise V_m as symmetric tensors in $(\mathbb{C}^2)^{\otimes m}$. Let $e_1^{m-k}e_2^k$ be the average of the $\binom{m}{k}$ words with $m-k$ letters e_1 and k letters e_2. The group acts by $g^{\otimes m}$, and the tensor inner product gives

$$\langle e_1^{m-k}e_2^k, e_1^{m-j}e_2^j\rangle = \delta_{jk}\binom{m}{k}^{-1}.$$

By the Peter–Weyl theorem, Theorem 10.11,

$$L^2(\mathrm{SU}(2)) = \widehat{\bigoplus}_{m\geq 0} E_m, \qquad \dim E_m = (m+1)^2 = 1, 4, 9, 16, \ldots$$

Under R, V_m occurs $m+1$ times. In physics, the matrix coefficients of V_m are called Wigner D-functions.

The block $m = 1$. For $V_1 = \mathbb{C}^2$, $c_{e_i,e_j}(g) = g_{ij}$. So $E_1 = \operatorname{span}\{a, -\bar{b}, b, \bar{a}\}$, and the functions $\sqrt{2}\, g_{ij}$ are orthonormal. In particular $\int |a|^2\, dg = \frac{1}{2}$. Under R, the row spans $\operatorname{span}\{a, -\bar{b}\}$ and $\operatorname{span}\{b, \bar{a}\}$ are copies of V_1.

The block m. Take the unit vector $u = e_1^m$. Since $ge_1 = ae_1 + be_2$ and the vectors $e_1^{m-k}e_2^k$ are orthogonal, only the $k = 0$ term of the binomial expansion has nonzero inner product with u:

$$c_{u,u}(g) = \left\langle e_1^m, (ae_1 + be_2)^m \right\rangle = a^m.$$

Then Schur orthogonality (10.1) gives $\int |a|^{2m}\, dg = 1/(m+1)$, matching $m = 1$.

Self-checks

1. Why is the operator (10.2) nonzero, although no invariant vector is known?

Answer. By the definition (10.2), $\langle u, Tu\rangle = \int |\langle \Pi(g)u, u\rangle|^2\, dg$. The integrand is continuous, nonnegative and equal to 1 at e. By positivity (H2), the integral is positive. Hence T is nonzero.

2. Give $\dim E_2 \subset L^2(\mathrm{SU}(2))$ and the multiplicity of V_2 under R.

Answer. The block has dimension $\dim E_2 = 9$. The representation V_2 occurs 3 times. Under L, $E_2 \cong \overline{V}_2^{\oplus 3}$.

3. Why does the coefficient proposition, Proposition 10.10, prove continuity before evaluating at e?

Answer. Only continuous elements of $L^2(K)$ have values. Evaluation is not even bounded on $\mathscr{R}(\mathrm{U}(1))$. For example, the function $f_N = (2N+1)^{-1/2} \sum_{|n|\le N} \chi_n$ has $\|f_N\|_2 = 1$ and $f_N(1) = \sqrt{2N+1}$.

Summary

Let K be a compact Lie group. In a strongly continuous unitary representation of K on $H \neq 0$, an averaged rank-one projection is a nonzero positive compact intertwiner. Its eigenspaces for nonzero eigenvalues are finite-dimensional and invariant. The finite-dimensional invariant subspaces span a dense subspace. In $L^2(K)$, smoothing identifies the finite-dimensional R-invariant subspaces with spaces of matrix coefficients. Schur orthogonality then gives the Peter–Weyl theorem. Under R, and also under L, each $\Pi \in \mathrm{Irr}(K)$ occurs in $L^2(K)$ with multiplicity d_Π. Density in $L^2(K)$ shows that $\mathscr{R}(K)$ separates points. By the Stone–Weierstrass theorem, $\mathscr{R}(K)$ is therefore uniformly dense in $C(K)$. Adding finitely many representations that separate points gives a faithful finite-dimensional unitary representation, so K is isomorphic as a Lie group to a closed subgroup of some $\mathrm{U}(n)$.

Reading

- Sepanski, *Compact Lie Groups*, §§3.2–3.3, including Theorem 3.25.
- Bröcker–tom Dieck, *Representations of Compact Lie Groups*, Chapter III, Sections 1–4.
- Etingof, MIT 18.755 lecture notes, Lectures 9–10.